Radially Extended Protoplanetary Disk Winds:
~~Non~~-Ideal MHD & Thermal Structure

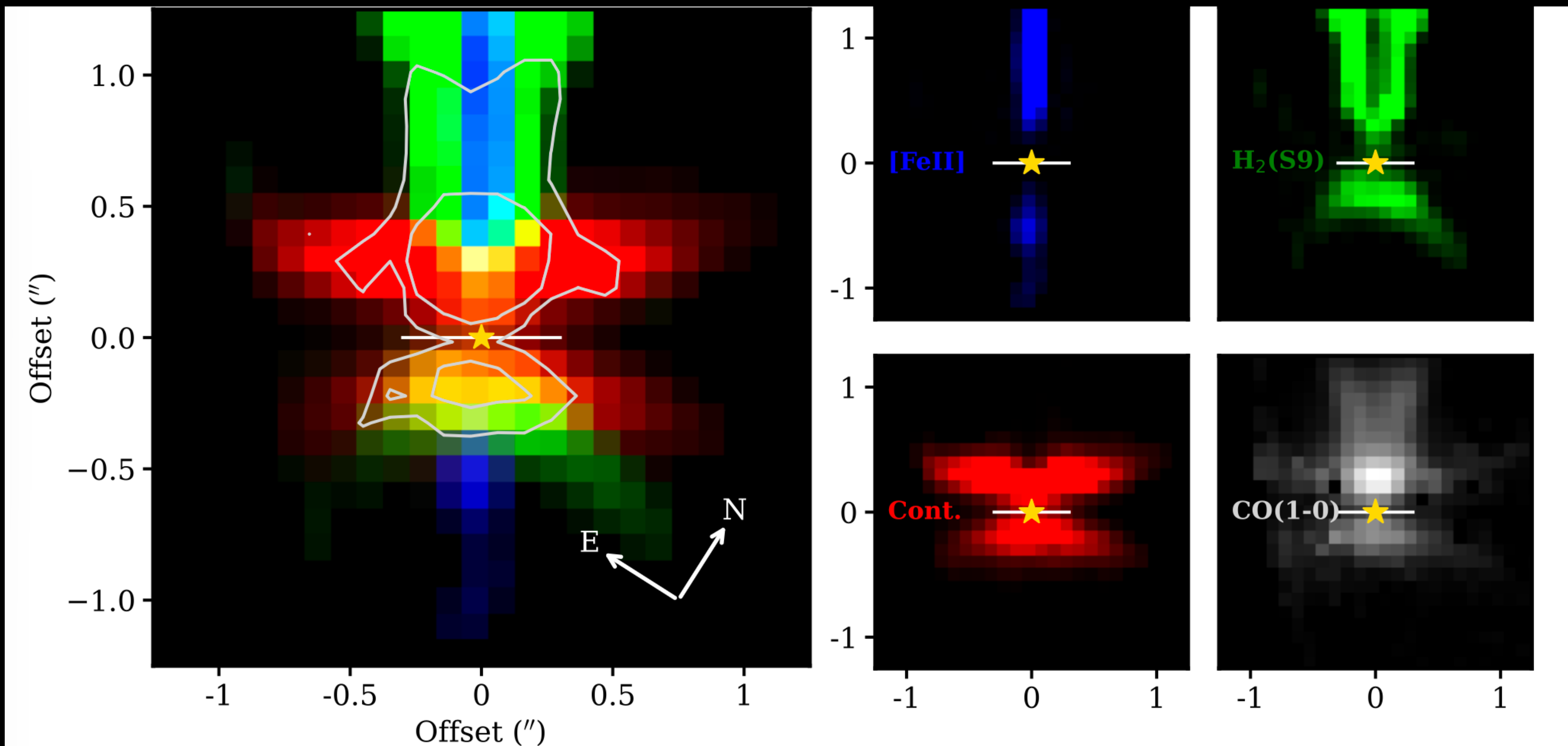
Fulvia Pucci, Uma Gorti, Neal Turner

(SETI)

Role of winds in accretion and in removing mass and angular momentum

- Turbulence
- Magnetic Fields – MRI
- Winds with ideal MHD
- Winds with non-ideal MHD (weak ionization)
- Thermal Pressure (better temperature profile)

HH 30 outflow JWST NIRSPEC images in H_2 S(9) and CO_{vib}



Pascucci+2025

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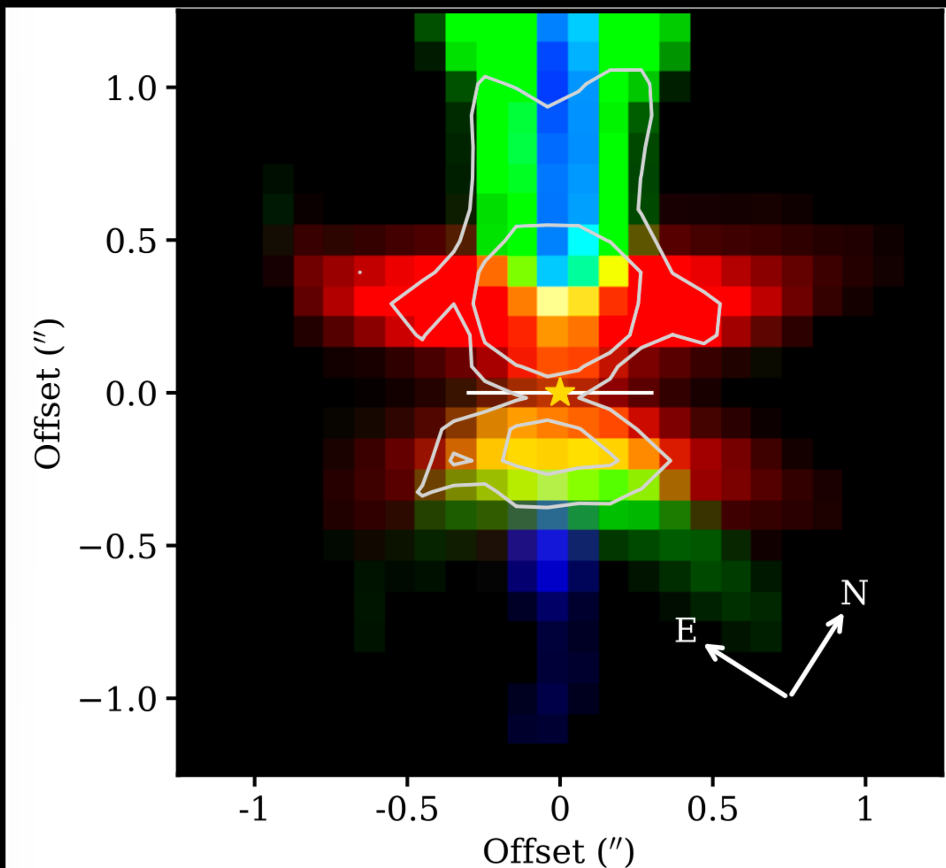
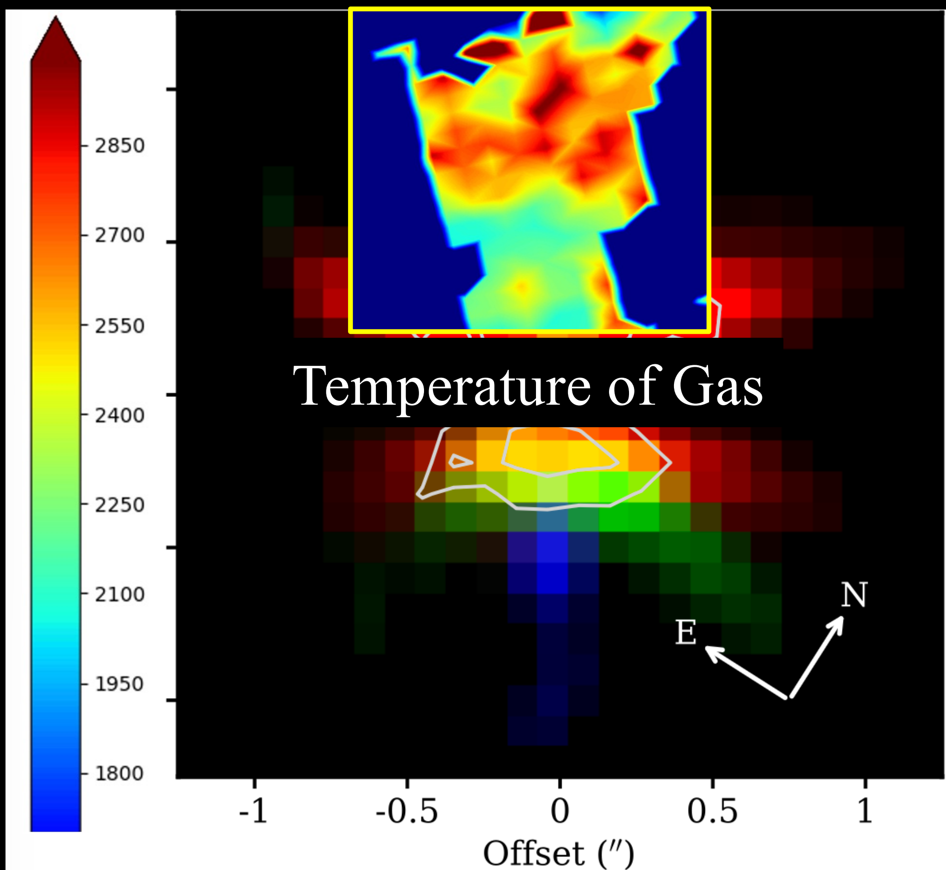


100's of au in scale

Pascucci+2025

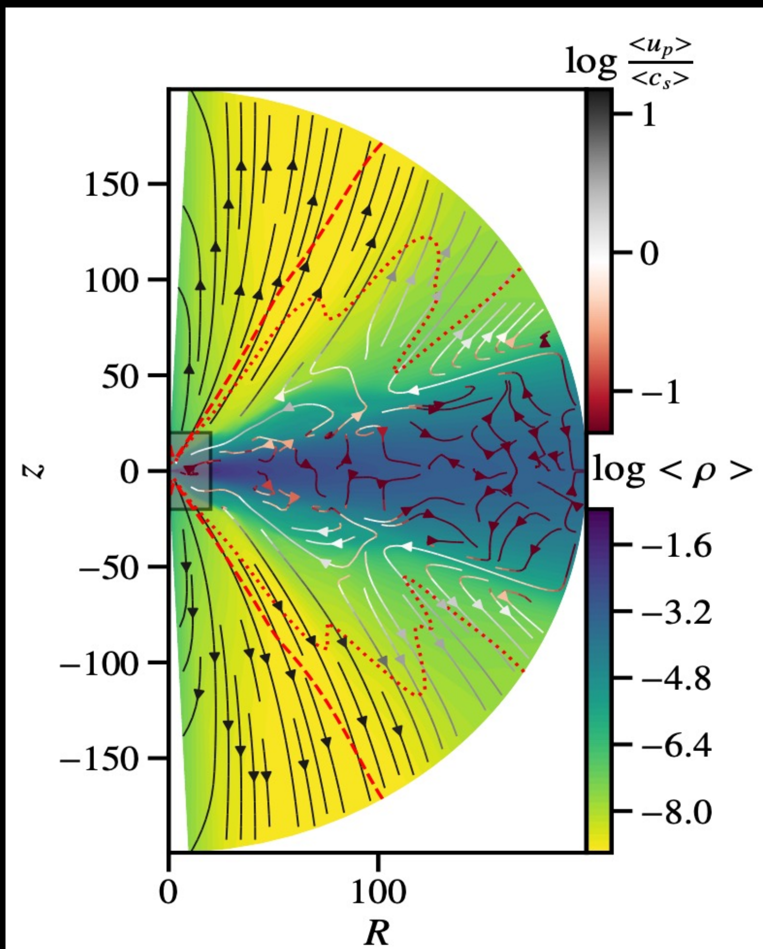
HH 30 outflow JWST NIRSPEC images in H_2 S(9) and CO_{vib}



100's of au in scale

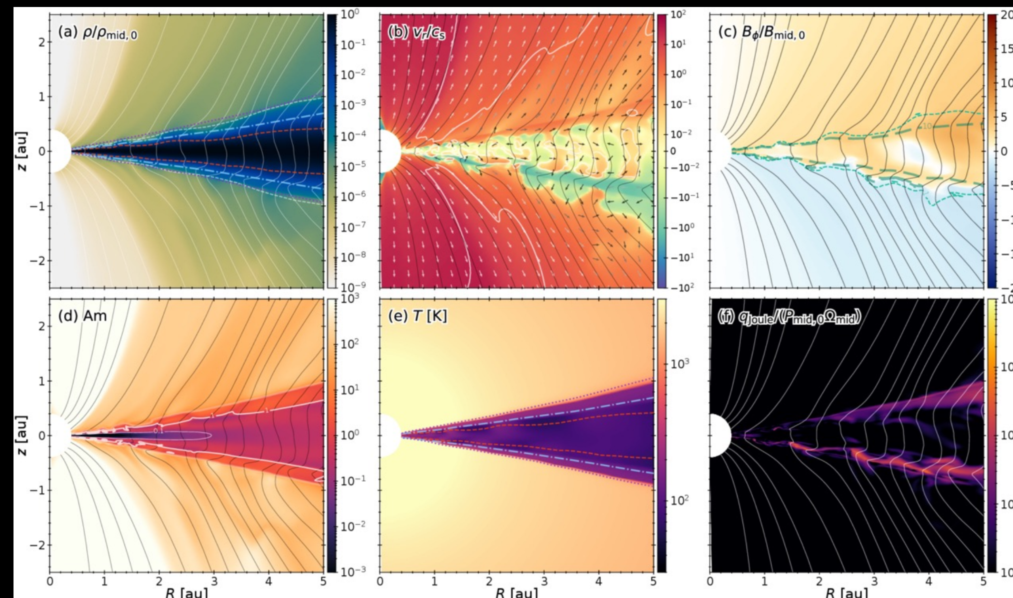
High excitation lines sensitive to T_{gas}

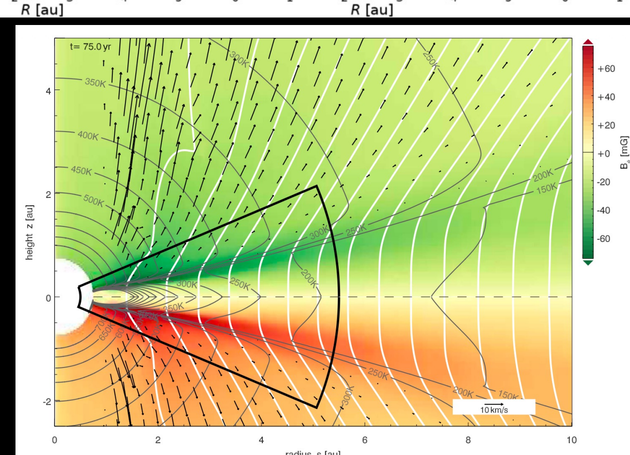
Pascucci+2025



Jacquemin-Ide & Lesur 2021

Mori, Bai & Tomida 2025





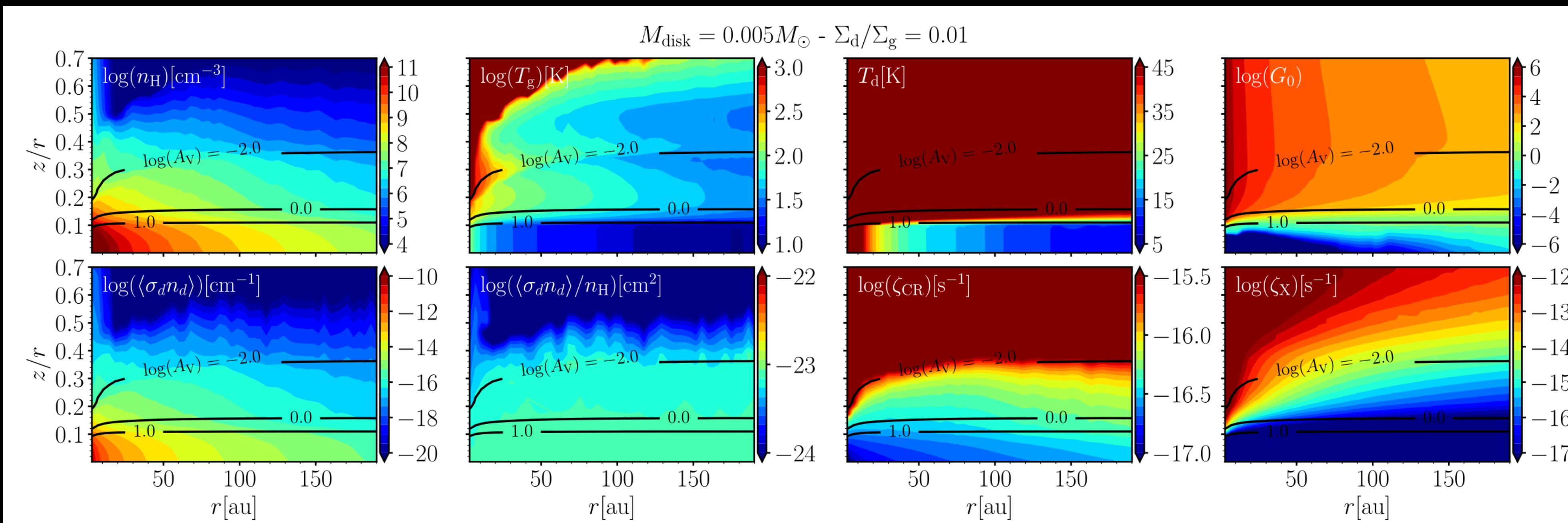
Gressel+2020

1. MHD simulations of winds on extended grid

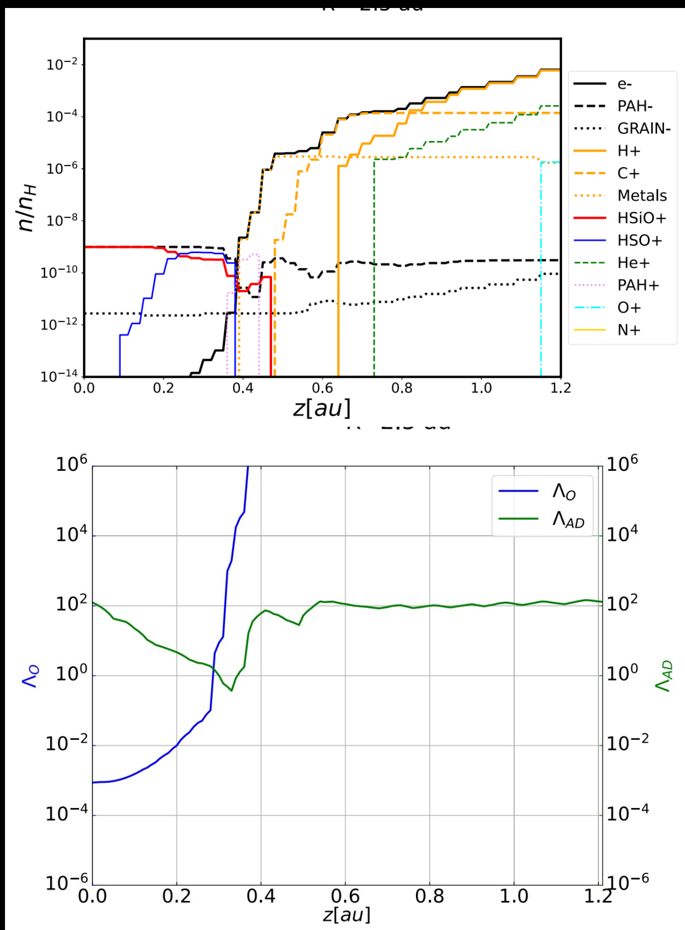
2. Temperature structure of disk from disk chemical models

Disk Gas and Dust temperatures can be very different

...very different structure, properties

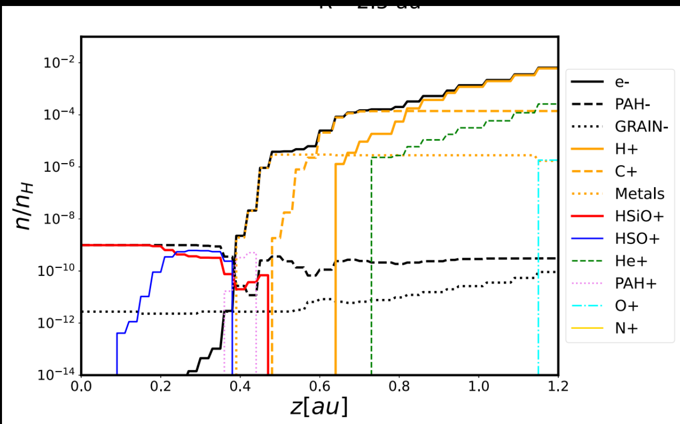


Ruaud & Gorti 2019



n/n_H
$z[au]$
e-
PAH-
GRAIN-
H+
C+
Metals
HSiO+
HSO+
He+
PAH+
O+
N+
Λ_O
Λ_{AD}
$z[au]$

Pucci+2024



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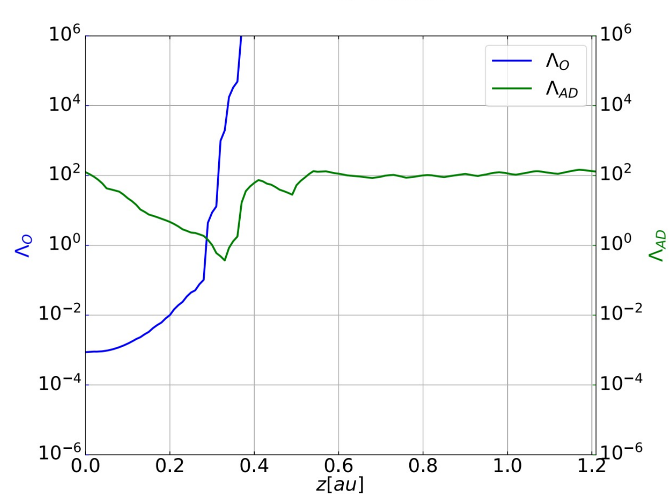
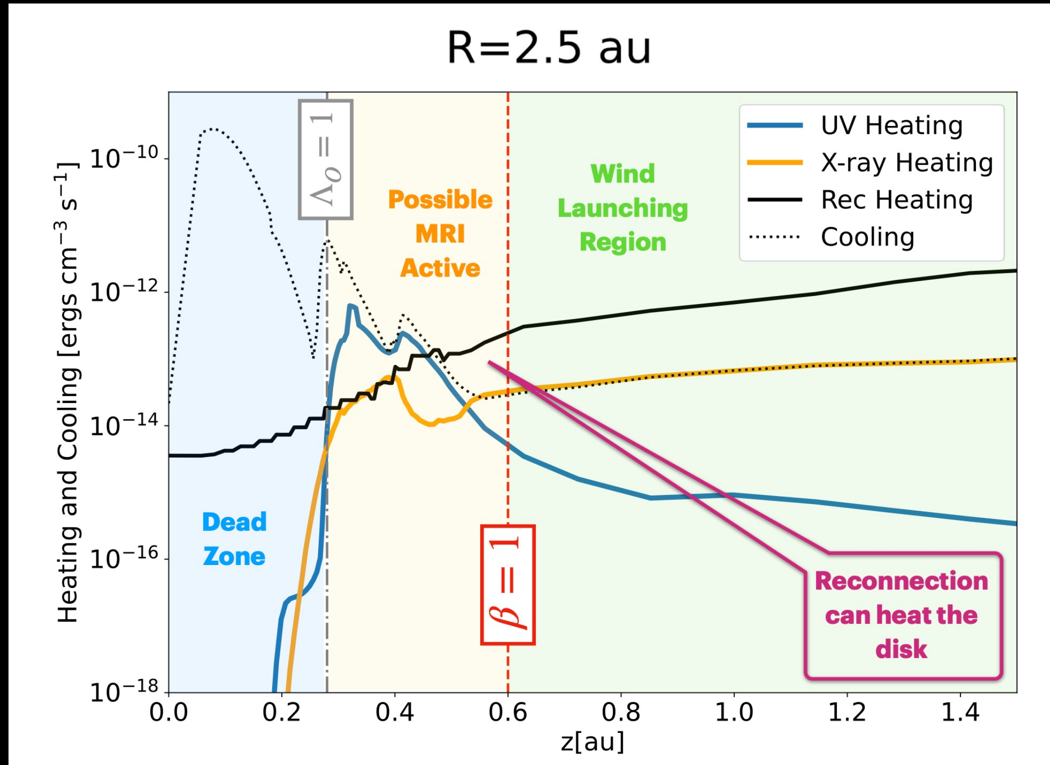


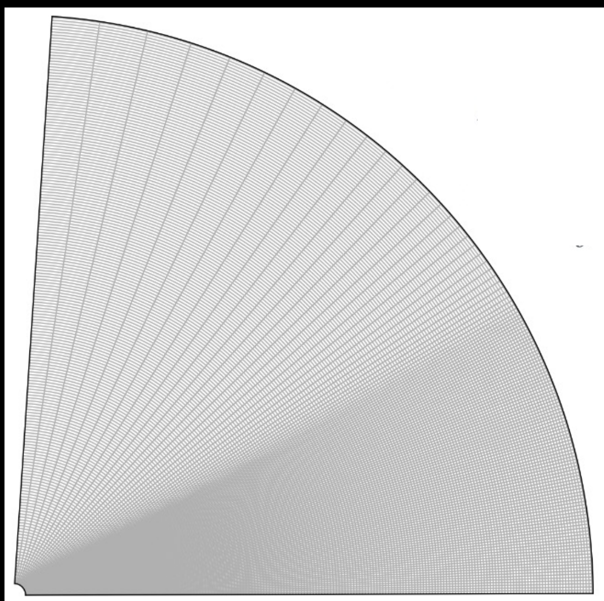
Λ_O
Λ_{AD}
$z[au]$



R=2.5 au
Heating and Cooling [ergs cm^{-3} s^{-1}]
z[au]
UV Heating
X-ray Heating
Rec Heating
Cooling
$\Lambda_O = 1$
$\beta = 1$
Dead Zone
Possible MRI Active
Wind Launching Region
Reconnection can heat the disk

Pucci+2024

Broad goal → Mass inflow and outflow rates



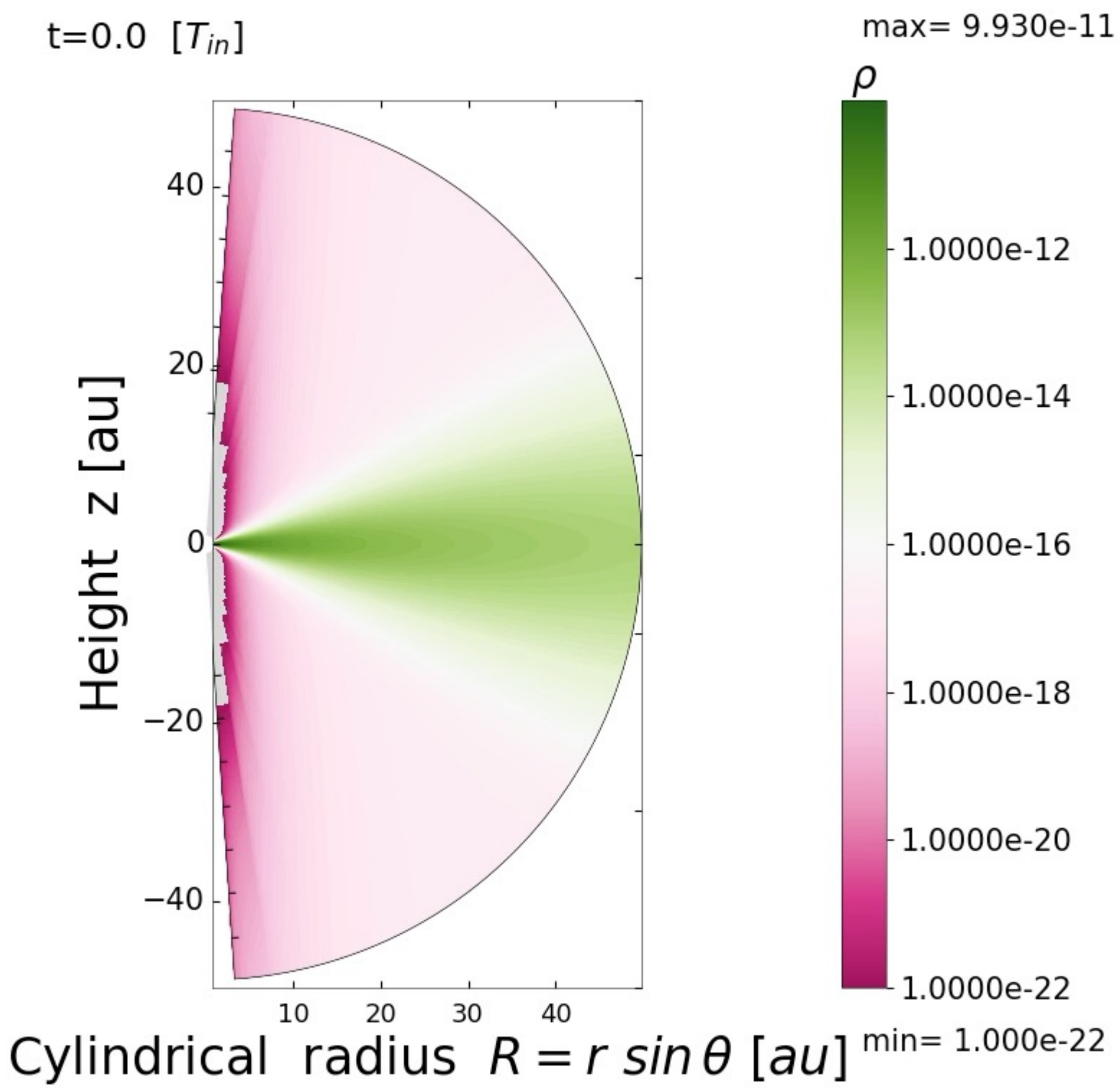
Athena++
r = [1, 50] au ; ~ 864
ϑ = [0, π] ; ~ 200
ϕ = [0, 2π] ; ~ 60

Systematic exploration:

1. 2D exploration, few 3D models where interesting
2. Set up disk in hydrostatic equilibrium (HSE)
3. Evolve to steady state, turn on B field
4. Isothermal disk, ideal MHD
5. Temperature profile, HSE, ideal MHD
6. Introduce non-ideal MHD

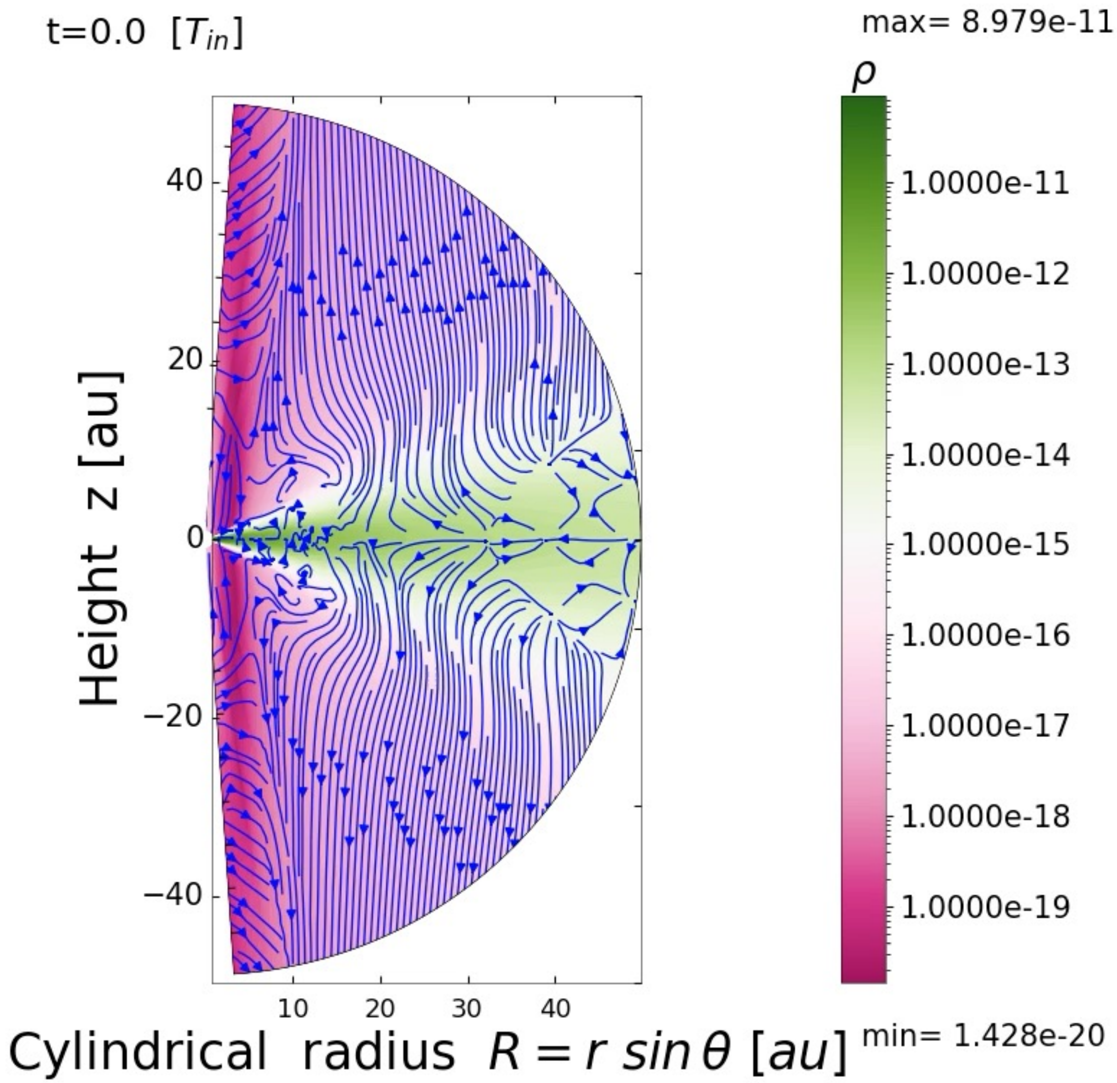
Pure Hydrodynamical Simulation

- Follow Jacquemin-Ide+Lesur (2021)
- Our density floor is 10^3 or more lower
- "Relax" to the temperature over ~ $0.1t_K$ to kill VSI
- Run sim longer to reach steady state thermal wind

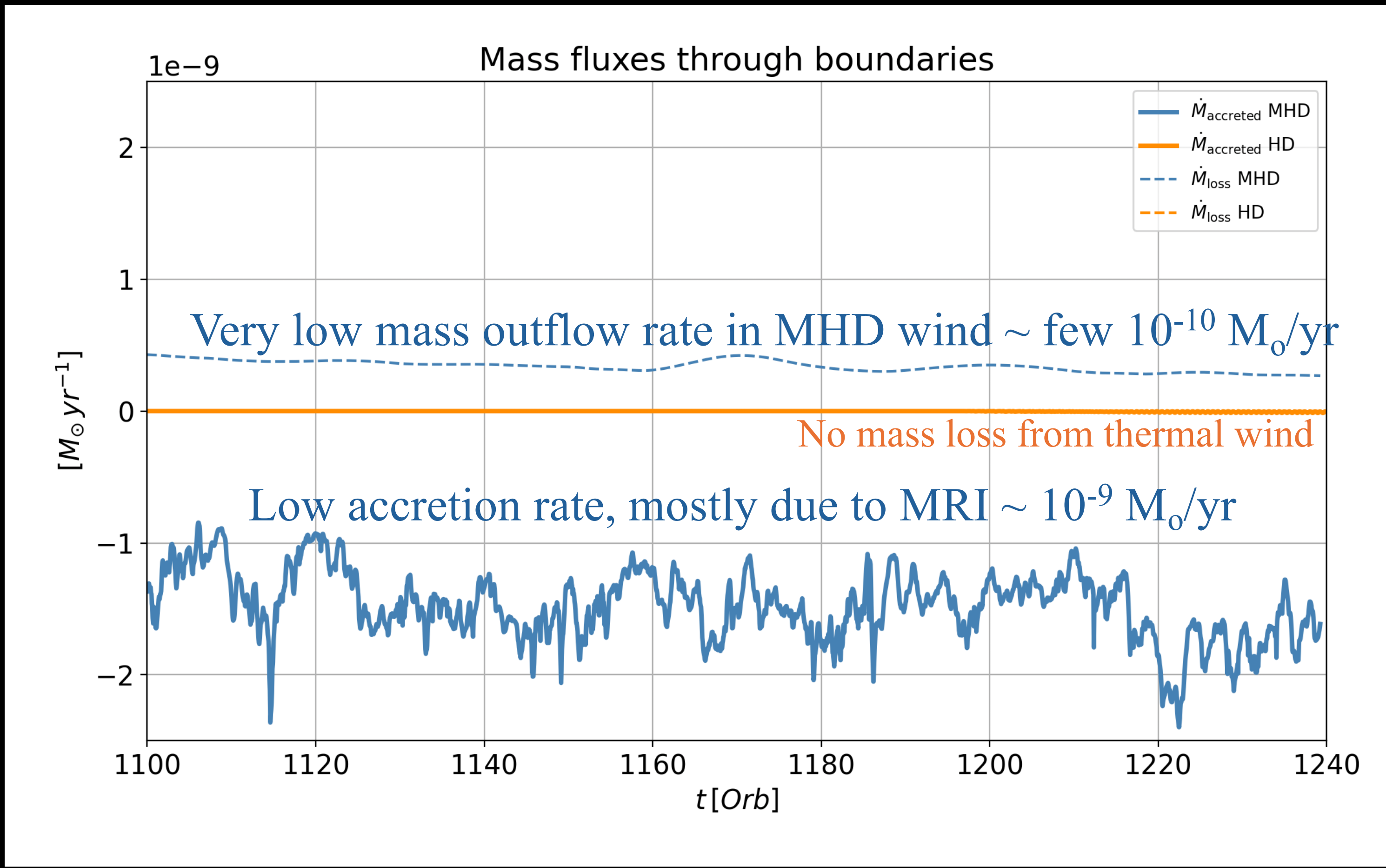


Magnetohydrodynamical Simulation

- Turn on B field after about ~ 1000 inner orbital times
- Limit vA to aid integration (J-I & L 2021)
- Different disk structure
- Simulation differs in that the plasma β is much lower (10^6 vs their 10^4), and we have a lower density floor



Isothermal vertical profile, qualitatively similar results as J-I & L 2021



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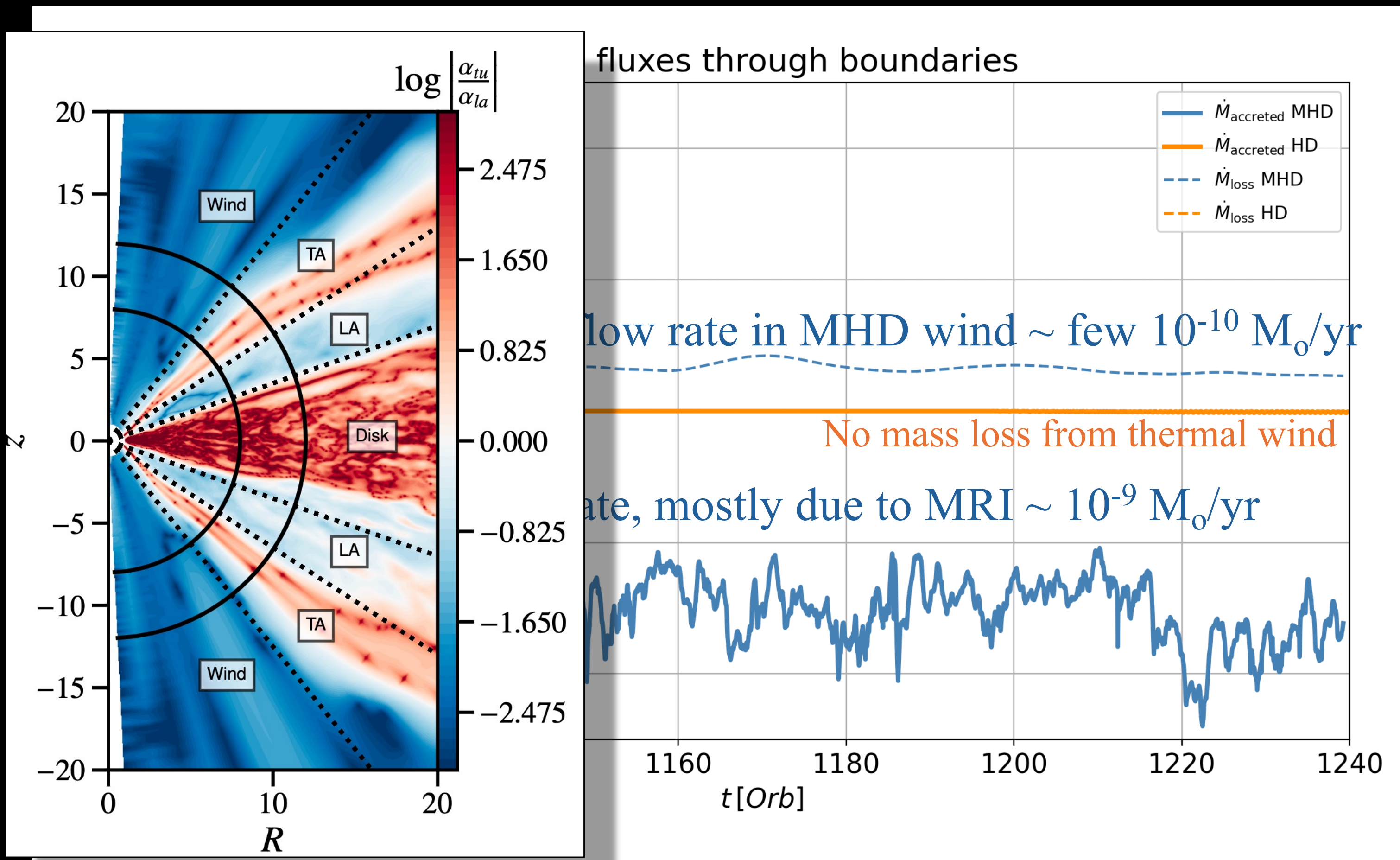
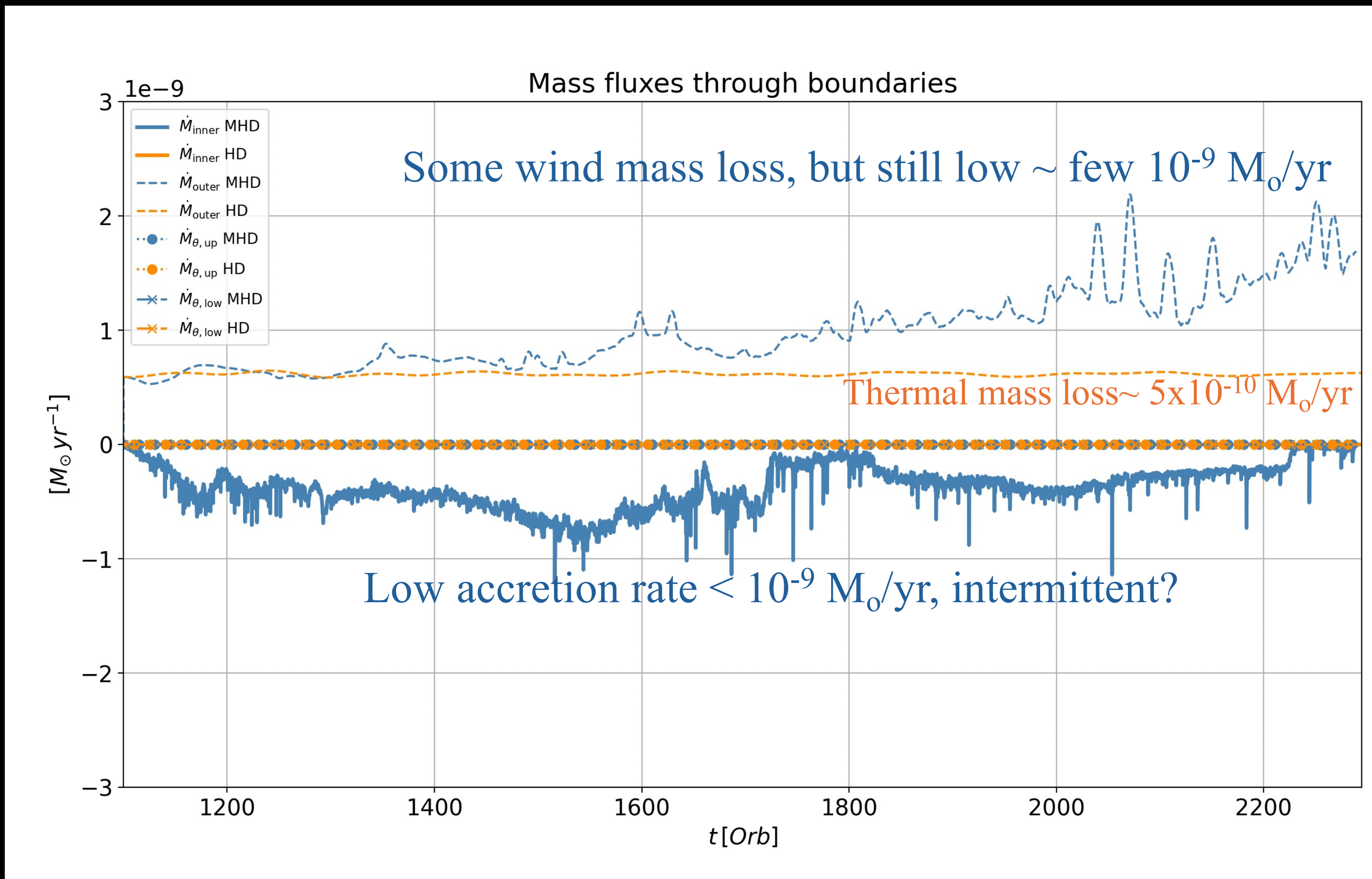


Realistic gas temperature (not evolved), ideal MHD



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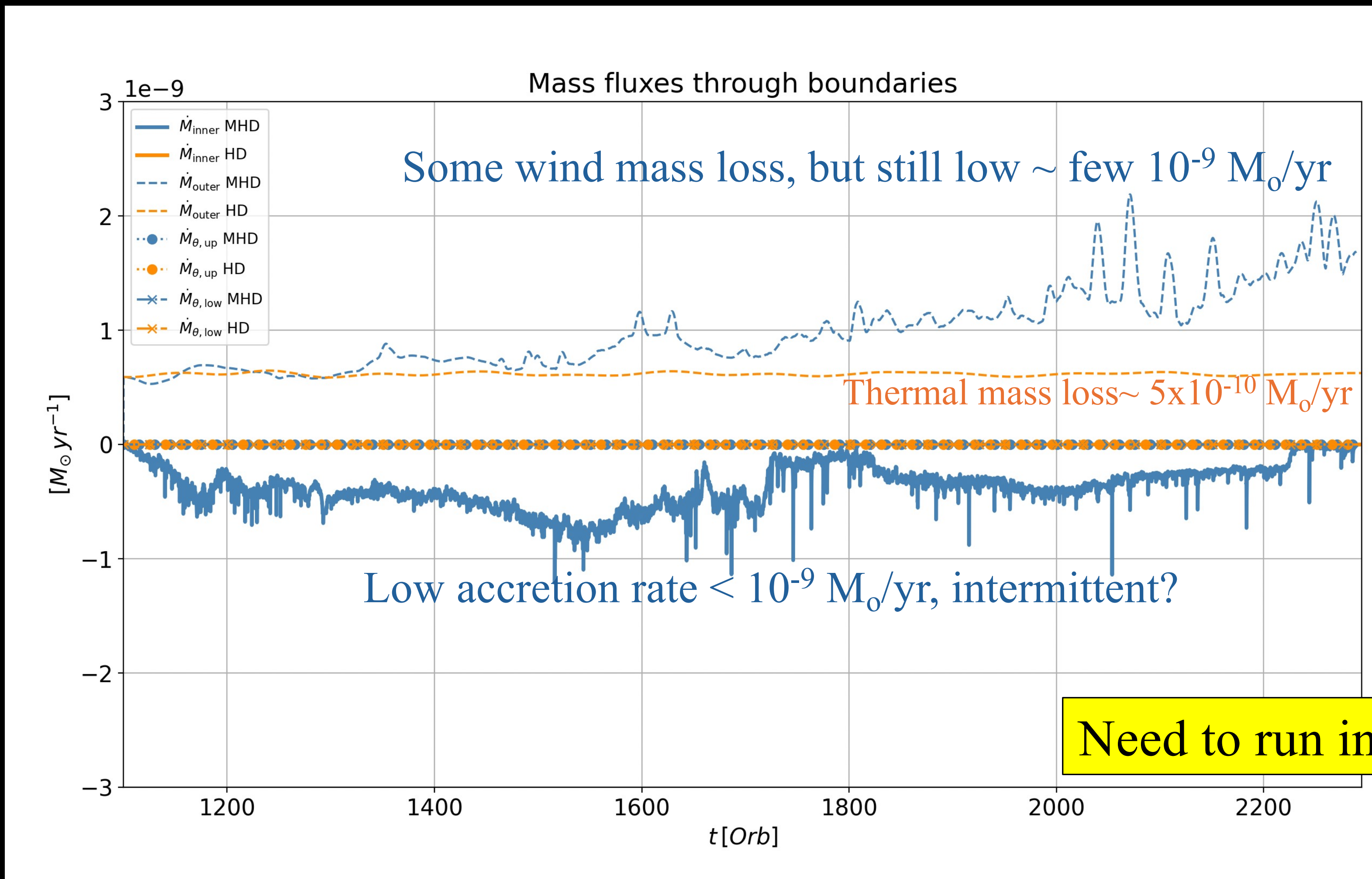

Need to run in 3D..

Summary: $\longrightarrow$ Mass inflow and outflow rates

With Ideal MHD and a realistic temperature,

- Lower mass accretion rate
- Low outflow rates as well, but $\dot{M}_{wind} > \dot{M}_{acc}$
- 3D ongoing – more MRI, surface layer accretion

(Pucci, Gorti & Turner, in prep)